MTH 200 **Skills Check 3 – Practice**

This skills check covers pre-requisite material needed for the third unit of the course, applications of derivatives. You should practice these ideas to ensure that you can quickly and correct answer such questions. Being able to do so will make learning the *calculus* material of the third unit much easier. It's never fun to be held up by Algebra/Trig. when trying to learn Calculus! *This is just a sample for practice. You may receive help on these questions and work with other students*.

1. **Factoring Fractional Exponents.**

(a) Given $x^{-2/3}\left(1+x^2\right)-2x^{1/3}$, factor out $x^{-2/3}$ and write your final answer using only positive exponents.

$$x^{-2/3}\left(1+x^2\right)-2x^{1/3} \Leftrightarrow x^{-2/3}\left[\left(1+x^2\right)-2x\right] \Leftrightarrow \frac{x^2-2x+1}{x^{2/3}}$$

(b) Given $2x\sqrt{2x+5}-x^2\left(2x+5\right)^{3/2}$, factor out $\left(2x+5\right)^{-1/2}$ and write your final answer using only positive exponents.

$$2x\sqrt{2x+5}-x^2\left(2x+5\right)^{3/2} \Leftrightarrow \left(2x+5\right)^{-1/2}\left[2x\left(2x+5\right)-x^2\left(2x+5\right)^2\right] \Leftrightarrow \frac{2x\left(2x+5\right)-x^2\left(2x+5\right)^2}{\left(2x+5\right)^{1/2}}$$

(c) Given $\left(1-x^2\right)^{-1/4}+2x\left(1-x^2\right)^{3/4}$, factor out $\left(1-x^2\right)^{-1/4}$ and write your final answer using only positive exponents.

$$\left(1-x^2\right)^{-1/4}+2x\left(1-x^2\right)^{3/4} \Leftrightarrow \left(1-x^2\right)^{-1/4}\left[1+2x\left(1-x^2\right)\right] \Leftrightarrow \frac{1+2x\left(1-x^2\right)}{\left(1-x^2\right)^{1/4}}$$

2. **Solving inequalities**

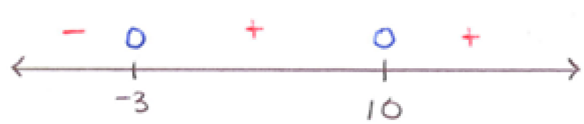
(a) Solve the inequality $\left(x+3\right)\left(x^2+9\right)\left(x-10\right)^2 \le 0$. Express you answer in interval notation.

We know that $\left(x+3\right)\left(x^2+9\right)\left(x-10\right)^2=0$ when $x=-3$ *or* $x=10$. Therefore, we put these values on a number line (representing the *x*-axis) and we check to see the behavior of the expression $\left(x+3\right)\left(x^2+9\right)\left(x-10\right)^2$ on the intervals on either side (and in between) these values.

If $x=-4$ then $\left(x+3\right)\left(x^2+9\right)\left(x-10\right)^2<0$

If $x=0$ then $\left(x+3\right)\left(x^2+9\right)\left(x-10\right)^2>0$

If $x=11$ then $\left(x+3\right)\left(x^2+9\right)\left(x-10\right)^2>0$



Since we are interested in when the expression is less than (or equal to) zero we have the solution set:

$$\left(-\infty,-3\right]\cup\left\{10\right\}$$

(b) Solve the inequality $\frac{e^x x^2}{x^2-25} \geq 0$. Express you answer in interval notation.

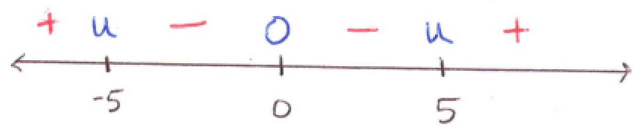
We know that $\frac{e^x x^2}{x^2-25} = 0$ when $e^x x^2 = 0 \iff x = 0$

We know that $\frac{e^x x^2}{x^2-25}$ is undefined when $x^2 - 25 = 0 \iff x = \pm 5$

Therefore, we put these values on a number line (representing the x-axis) and we check to see the behavior of the expression $\frac{e^x x^2}{x^2-25}$ on the intervals on either side (and in between) these values.

If $x = -6$ then $\frac{e^x x^2}{x^2-25} > 0$

If $x = -1$ then $\frac{e^x x^2}{x^2-25} < 0$



If $x = 1$ then $\frac{e^x x^2}{x^2-25} < 0$

If $x = 6$ then $\frac{e^x x^2}{x^2-25} > 0$

Since we are interested in when the expression is greater than (or equal to) zero we have the solution set:

$$(-\infty,-5)\cup\{0\}\cup(5,\infty)$$

(c) Solve the inequality $\frac{6x^2+5x+1}{(x-4)} < 0$. Express you answer in interval notation.

We factor the expression to find: $\frac{6x^2+5x+1}{(x-4)} = \frac{(2x+1)(3x+1)}{(x-4)}$.

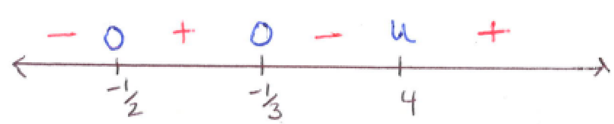
We know that $\frac{(2x+1)(3x+1)}{(x-4)} = 0$ when $(2x+1)(3x+1) = 0 \iff x = -\frac{1}{2}$ *or* $-\frac{1}{3}$

We know that $\frac{(2x+1)(3x+1)}{(x-4)}$ is undefined when $x - 4 = 0 \iff x = 4$

Therefore, we put these values on a number line (representing the x-axis) and we check to see the behavior of the expression $\frac{(2x+1)(3x+1)}{(x-4)}$ on the intervals on either side (and in between) these values.

If $x = -1$ then $\frac{(2x+1)(3x+1)}{(x-4)} < 0$

If $x = -0.4$ then $\frac{(2x+1)(3x+1)}{(x-4)} > 0$



If $x = 0$ then $\frac{(2x+1)(3x+1)}{(x-4)} < 0$

If $x=5$ then $\frac{(2x+1)(3x+1)}{(x-4)}>0$

Since we are interested in when the expression is less than zero we have the solution set:

$\left(-\infty,-\frac{1}{2}\right)\cup\left(-\frac{1}{3},4\right)$

3. Solving Exponential and Logarithmic Equations

(a) Solve the equation $3xe^x-2e^x=0$

$$e^x(3x-2)=0 \quad\Leftrightarrow\quad (3x-2)=0 \quad\Leftrightarrow\quad x=\frac{2}{3}$$

(b) Solve the equation $e^{2x}-e^x=0$

$$e^x\left(e^x-1\right)=0 \quad\Leftrightarrow\quad e^x-1=0 \quad\Leftrightarrow\quad e^x=1 \quad\Leftrightarrow\quad x=0$$

(c) Solve the equation $x\ln(x)=x$

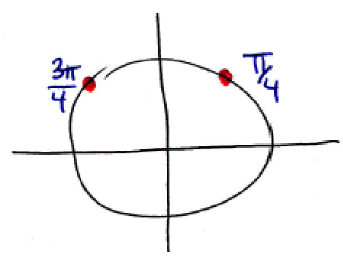
$$x\ln(x)-x=0 \quad\Leftrightarrow\quad x\left[\ln(x)-1\right]=0 \quad\Leftrightarrow\quad x=0 \ or\ \ln(x)-1=0$$

$$\Leftrightarrow\quad x=0 \ or\ \ln(x)=1 \quad\Leftrightarrow\quad x=0 \ or\ x=e$$

4. Solving Trigonometric Equations

(a) Solve the equation $2\sin(\theta)=\sqrt{2}$ for θ.

$$2\sin(\theta)=\sqrt{2} \quad\Leftrightarrow\quad \sin(\theta)=\frac{\sqrt{2}}{2}$$

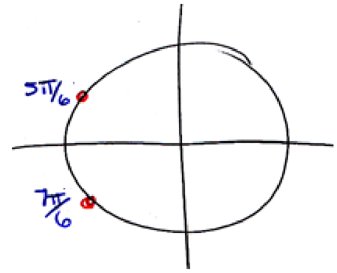


$\theta=\frac{\pi}{4}+2\pi k \ \ or\ \ \theta=\frac{3\pi}{4}+2\pi k$, where k is an integer.

(b) Solve the equation $\sqrt{3}\cos(\theta)=-\frac{3}{2}$ for θ.

$$\sqrt{3}\cos(\theta)=-\frac{3}{2} \quad\Leftrightarrow\quad \cos(\theta)=-\frac{3}{2\sqrt{3}} \quad\Leftrightarrow\quad \cos(\theta)=-\frac{3}{2\sqrt{3}}\cdot\frac{\sqrt{3}}{\sqrt{3}} \quad\Leftrightarrow\quad \cos(\theta)=-\frac{\sqrt{3}}{2}$$

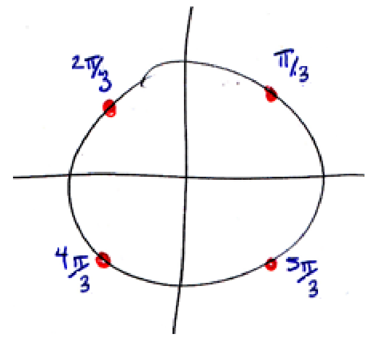
$$\theta = \frac{5\pi}{6} + 2\pi k \;\; or \;\; \theta = \frac{7\pi}{6} + 2\pi k$$, where k is an integer.



(c) $\tan^2(\theta) - 3 = 0$

$$\tan^2(\theta) - 3 = 0 \;\; \Leftrightarrow \;\; \tan(\theta) = \pm\sqrt{3}$$

$$\theta = \frac{\pi}{3} + \pi k \;\; or \;\; \theta = \frac{2\pi}{3} + \pi k$$, where k is an integer.



5. Formulas for Areas, Volumes, and Dimensions of Shapes

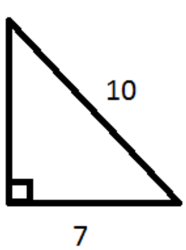
(a) Find the area of a circle with diameter $\sqrt{3}$.

$$A = \pi r^2 = \pi \left(\frac{\sqrt{3}}{2} \right)^2 = \frac{3\pi}{4}$$

(b) Find the volume of a cylinder with a height of 1 meter and a diameter of 1 meters.

$$V = \pi r^2 h = \pi \left(\frac{1}{2} \right)^2 (1) = \frac{\pi}{4}$$

(c) Determine the length of the remaining side of the triangle below.



$$x^2 + 7^2 = 10^2 \;\; \Leftrightarrow \;\; x = \sqrt{100 - 49} = \sqrt{51}$$